

EXECUTIVE SUMMARY

Product Line Profitability & Margin Performance Analysis for Nassau Candy Distributor

This project, **“Product Line Profitability & Margin Performance Analysis for Nassau Candy Distributor,”** focuses on evaluating the financial performance of different product lines using data analytics and interactive dashboard visualization techniques. The study addresses a critical business challenge where high sales volumes do not necessarily translate into high profitability. Many products generate significant revenue but contribute lower profit margins due to operational costs, manufacturing expenses, discounts, and supply chain inefficiencies. The project aims to identify the products and divisions that truly drive profitability and support more effective business decision-making.

The analysis was conducted using tools such as Python, Pandas, Plotly, and Streamlit. The dataset consisted of 10,194 records containing information related to sales, cost, gross profit, divisions, products, customers, and shipment details. Data cleaning and preprocessing techniques were applied to handle missing values, remove duplicates, and transform raw data into meaningful analytical metrics. Key performance indicators such as Gross Margin Percentage, Profit per Unit, Revenue Contribution, and Profit Contribution were calculated to evaluate product and division performance.

The project also involved the development of an interactive dashboard using Streamlit, enabling dynamic filtering, profitability tracking, and visualization of business insights through charts, scatterplots, Pareto analysis, and geographical mapping. The findings revealed that the Chocolate division contributed the highest share of sales and profit, while a small number of products generated the majority of overall profitability. Additionally, some low-margin products were identified as potential candidates for discontinuation or strategic review.

Overall, the study demonstrates the importance of profitability-focused analysis and data-driven decision-making in modern business operations. The dashboard provides management with actionable insights for improving pricing strategies, optimizing operational efficiency, and enhancing long-term business sustainability.